

Fundamental Group

1.1 Exercises

Problem 1.1.1 ► Mobius band

Let M be the Mobius band, that is, the quotient of $[0, 1]^2$ under the equivalence relation generated by $(x, 0) \sim (1 - x, 1)$ for all $x \in [0, 1]$. Prove that M is homotopy equivalent to $\mathbb{S}^1$.

Proof. Construct

$$f : [0, 1]^2 \times [0, 1] \rightarrow \mathbb{S}^1$$

$$(x, y, t) \mapsto \left(\frac{1}{2}t + (1 - t)x, y \right)$$

Denote $f_t(x, y) = f(x, y, t)$, then

$$f_0(x, y) = (x, y), \quad f_1(x, y) = \left(\frac{1}{2}, y \right), \quad f_t\left(\frac{1}{2}, y\right) \equiv \left(\frac{1}{2}, y\right)$$

Which is a SDR to $\mathbb{S}^1$. □

Problem 1.1.2

1. Prove that the inclusion $O_n(\mathbb{R}) \rightarrow GL_n(\mathbb{R})$ is a homotopy equivalence. (Use the Gram-Schmidt orthonormalization procedure to construct the reverse map).
2. Among the following matrix groups, determine which ones are compact, and determine their π_0 :

$$GL_n(\mathbb{C}), GL_n(\mathbb{R}), O_n(\mathbb{R}), SO_n(\mathbb{R}), U_n(\mathbb{C}), SU_n(\mathbb{C}).$$

Proof. Take $A \in \mathrm{GL}(n, \mathbb{R})$. Let $A = (v_1, \dots, v_n)$. Let

$$\begin{aligned} e_1 &= \frac{v_1}{\|v_1\|} \\ e_2 &= \frac{v_2 - \langle v_2, e_1 \rangle e_1}{\|v_2 - \langle v_2, e_1 \rangle e_1\|} \\ &\dots \end{aligned}$$

$\det Q \det(t \mathrm{Id} + (1 - t)R) \neq 0$. Let $Q = (e_1, \dots, e_n)$. $A = QR$ for some upper triangular matrix R . Let

$$\begin{aligned} f_t &: \mathrm{GL}_n(\mathbb{R}) \rightarrow \mathrm{GL}_n(\mathbb{R}) \\ A &\mapsto Q(t \mathrm{Id} + (1 - t)R) \end{aligned}$$

which is a SDR from A to Q . □